Competitive Exam Practice 4: System Programs and OS Structures

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. *All questions in this set are original, written in the GATE pattern; none is claimed to be a past-year question.*

Q1 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which of the following is **NOT** a system program?

(A) The command interpreter (shell)

(B) A compiler shipped with the operating system

(C) The CPU scheduler

(D) A linkage editor

Q2 *[Original, GATE-pattern]* **[MCQ, 1 mark]** In a UNIX-style shell where most commands are implemented as separate system programs, which command must nevertheless be a shell built-in?

(A) `ls`

(B) `cd`

(C) `rm`

(D) `cp`

Q3 *[Original, GATE-pattern]* **[MCQ, 1 mark]** A timer interrupts the CPU after a fixed interval, and a configuration file sets the interval for each user class. Which statement is correct?

(A) Both the timer and the interval are policies

(B) The timer is a mechanism; the interval chosen for each class is a policy

(C) The timer is a policy; the interval is a mechanism

(D) Both are mechanisms, because both are implemented in the kernel

Q4 *[Original, GATE-pattern]* **[MCQ, 2 marks]** A device driver dereferences a `NULL` pointer. In which structure is the failure most likely to be confined to the driver alone, with the rest of the system continuing to run?

(A) Monolithic kernel

(B) Layered kernel

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(C) Microkernel with the driver as a user-level server

(D) Monolithic kernel with the driver as a loadable module

Q5 *[Original, GATE-pattern]* **[MCQ, 2 marks]** Consider the statements **S1:** in a layered OS, every layer above layer 0 runs in user mode; **S2:** a loadable kernel module runs in kernel mode.

(A) Only S1 is true

(B) Only S2 is true

(C) Both are true

(D) Neither is true

Q6 *[Original, GATE-pattern]* **[MCQ, 2 marks]** The principal disadvantage of the microkernel approach relative to a monolithic kernel is:

(A) it cannot support device drivers

(B) increased overhead from message passing and extra user/kernel crossings

(C) it cannot be ported to new hardware

(D) a bug in any service always crashes the whole system

Q7 *[Original, GATE-pattern]* **[NAT, 2 marks]** In a microkernel, a read request travels from an application through a chain of 3 user-level servers, and the replies return along the same chain. Each message is one trap, and each trap causes 2 user/kernel boundary crossings. The total number of boundary crossings for the request is _____.

Q8 *[Original, GATE-pattern]* **[NAT, 2 marks]** A layered OS has layers 0 (hardware) through 6 (user interface). A request entering at layer 6 calls each lower layer once on the way down and returns through each on the way up. Each inter-layer call or return costs 40 ns, and the useful work at layer 0 takes 1.2 μs. The percentage of the total request time spent on inter-layer overhead is _____ (round to one decimal place).

Q9 *[Original, GATE-pattern]* **[NAT, 2 marks]** In a layered design, the rule "if A calls B, then B is in a lower layer" is applied to four components F, M, D, S with calls F$\rightarrow$M, F$\rightarrow$D, M$\rightarrow$D, D$\rightarrow$S. The number of distinct valid orderings of the four components (each in its own layer) is _____.

Q10 *[Original, GATE-pattern]* **[NAT, 2 marks]** Each user/kernel crossing costs 0.5 μs. A monolithic kernel serves a file read with 1 trap; a microkernel serves it through a chain of 2 servers (count $2k$ traps for k servers, 2 crossings per trap). For 100,000 reads per second, the additional CPU time per second, in milliseconds, that the microkernel spends on crossings is _____.

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